AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 18 · TEACHER GUIDE

Confidence Is Not Correctness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Teach confidence as a claim requiring appropriate evidence.

Model score meanings, group outcomes and missing coverage, then reserve the leaf-sorter audit for independent evidence.

Keep one habit: Look for separate claim definitions, outcome checks and bounded conclusions.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Distinguish a **confidence claim** from **evidence** that a **prediction** was correct

Keep **confidence**, numerical scores and **correctness** separate. **Calibration** is introduced as a group comparison, without formal estimation. Small exact matches do not prove **calibration**.

Concept reference: scikit-learn probability **calibration** documentation at https://scikit-learn.org/1.8/modules/**calibration**.html. All classroom examples here are fictional.

By the end, I can explain why a **confidence** score is not a guarantee and account for missing outcomes.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Separate cards for probability claims and later outcomes
- Ten counters to show observed correct and incorrect cases
- W4 leaf-sorter records, with one outcome deliberately missing

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

Look for: Separate a recorded **prediction** from the **observation** used to check it

Reference example, not the only acceptable wording: **Input** height 6 cm; **prediction** far because 6 is at least 6; observed distance 3 units; observed category near because 3 is below 4; mismatch.

Height is available to the model before the trial, while travel distance is measured afterward to check its **prediction**.

Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

Look for: Separate examples used to choose a model from examples reserved for its final evaluation

Reference example, not the only acceptable wording: Comparing candidates is development. Scoring a frozen model on previously reserved examples is evaluation, provided they did not influence its development.

Changing a threshold after reading the test score uses that test information for development of the revision; the same set is no longer independent of that revision.

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

Look for: Compare expected and observed traces to locate the first difference

Reference example, not the only acceptable wording: Expected trace: 3, 5, 10, 9. First difference is the doubling checkpoint: 10 expected versus 8 observed.

The later 8 minus 1 equals 7, so the observed subtraction is consistent with its **input**. Investigate the earlier step before claiming another fault.

No. The case is unassessed until adequate outcome **evidence** is available. The strength of the wording supplies no observed distance.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Prediction Is Not Fact

Save New Examples for Testing

Debug the Cause

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A helper says 'definitely correct' on four answers; only two match the answer key. A cautious helper gives the same four answers. Which is more accurate? What additional **evidence** would you need to compare **confidence** quality?

Concept to explain

A system can be very confident and still be wrong.

Term	Meaning
confidence	strength assigned to an output by a system
correctness	agreement with verified outcome
calibration	how well confidence matches observed accuracy
confident error	wrong result with strong stated confidence

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach confidence through definitions and checked groups

> Keep wording, score definitions, probability estimates, observed correctness and missing coverage separate.

Begin with the three meanings of a confidence display

A **prediction** can be expressed strongly and still be wrong. **Confidence** describes a claim about support or likelihood; **correctness** is checked against **evidence**.

Keep the **prediction**, the **confidence claim** and the observed outcome in separate columns. None can replace the others.

A display reading 80 might mean 80 out of 100 points for matching features, an estimated 80 percent chance of a specified event, or something else. Read its definition before interpreting it.

The number alone does not tell you which meaning applies. A sentence such as definitely far is not a measured probability.

Ramp1 from Week 17 produced categories using a threshold. It supplied no **probability estimate**s. We must not attach a probability merely because the height is far from that threshold.

This week uses fictional forecast records with an explicitly stated probability claim; we are checking those claims, not inventing them for Ramp1.

Explain group comparisons without exact-count guarantees

An 80 percent claim means an estimated chance of 80 out of 100, for the event that was named. Across many relevant cases carrying that claim, a well-calibrated method should be correct about that often.

It does not promise exactly eight successes in every ten cases or tell us which cases will fail.

Eight correct out of ten agrees numerically with 80 percent in this small group. Five out of ten is lower. Both are **observation**s worth reporting; neither small group alone establishes how dependable the method will be.

Larger relevant checks, independent of model **fitting** and score adjustment, give a stronger basis for judging the claim.

Model the selected ten-card evidence record

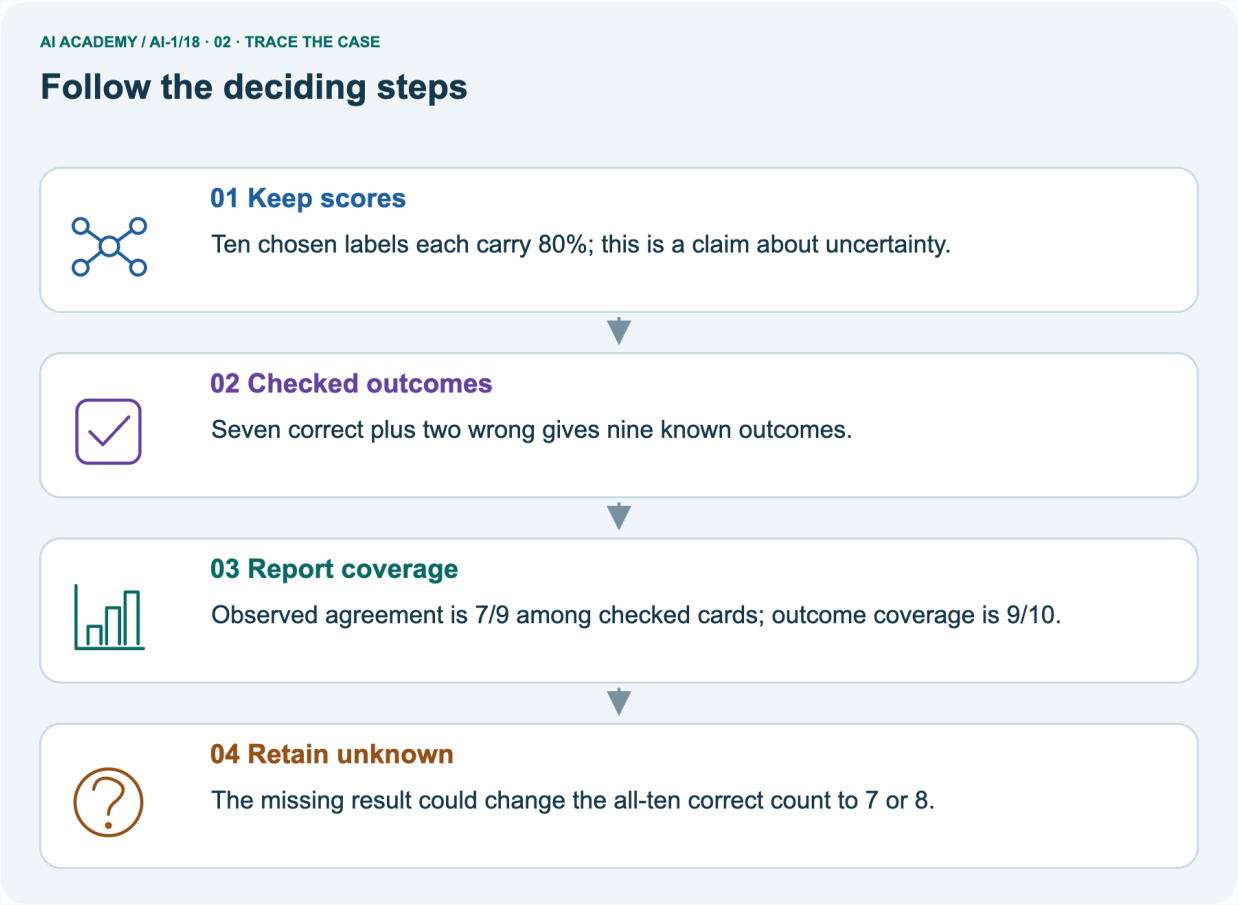
A fictional card classifier reports a claimed **80%** chance that each chosen label is correct.

Ten **prediction**s all have score **80%**.

Later checks find 7 correct, 2 incorrect and 1 outcome unavailable. The selected cards were not a representative survey.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Do not count the missing outcome as correct or silently remove it from the coverage report. These records allow an all-ten agreement range from **70%** to **80%**.

They do not show that claimed probabilities match checked **correctness** over many comparable **prediction**s; that requires much more suitable **evidence**.

Resolve the missing outcome as incorrect

1. Change: The missing outcome is checked and found incorrect.
2. **Prediction**: All-ten agreement is 7/10=**70%**.
3. Reason: There are now three wrong labels and no unknown outcomes.

Limit the calibration and future-use claim

This small selected set cannot establish that the claimed probabilities match actual **correctness** or show performance for all future cards.

Inspect individual high-confidence errors and unknowns

Case and recorded claim	Observed outcome	Judgment
A predicts far at 90 percent	Verified near	Incorrect despite high confidence
B predicts near at 60 percent	Verified near	Correct despite lower confidence
C predicts far at 90 percent	Outcome missing	Unassessed

A high-**confidence** error and a lower-**confidence** success can occur even when a probability method is useful. One error does not prove that all **confidence** estimates are worthless, and one success does not validate a method.

Keep the original scores and investigate patterns across relevant cases.

Record the task, model version, event being assigned a probability, score definition and **evidence** used for checking. If outcomes are missing, report that count.

Do not quietly remove difficult cases or change the **confidence** values after reading the answers. Adjustments belong to a new recorded method and need a fresh evaluation.

Use the picture deliberately

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT input and model scores

2. Process — the deciding step

PROCESS rank possible **output**s

3. Output — the result

OUTPUT predicted label plus **confidence**, later checked for **correctness**

CORE IDEA A system can be very confident and still be wrong.

UNDER THE HOOD **Confidence** is model-relative strength, not a calibrated probability of truth unless validated. Distribution shift, bad labels, missing classes, or over**fitting** can produce high-**confidence** errors.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE A classifier assigns 95 percent **confidence** to 'dog' for a toy fox but the verified label is fox.

STEP 1 — OBSERVE **input** and model scores

STEP 2 — TRACE rank possible **output**s

STEP 3 — CONCLUDE predicted label plus **confidence**, later checked for **correctness**

MODEL REASONING **Confidence** and **correctness** are separate axes. Only verified outcomes establish **correctness**. Compare many cases to learn whether **confidence** is trustworthy; one 95 percent value is not proof.

DO NOT OVERCLAIM Avoid presenting a score as an objective certainty when its meaning has not been validated.

What new **evidence** could change this answer?

Concrete teaching case

A helper confidently labels a blurred dog photo as cat; the clear original shows a dog.

What does this show?

Exact explanation to model

Confidence and **correctness** differ. A **confident error** remains wrong. Check **evidence**, particularly unfamiliar or low-quality **input**s.

Explain the added worked example

Key terms

Term	Meaning
Confidence claim	A statement or number expressing how strongly a prediction is supported according to a stated method
Correctness	Whether a prediction agrees with an adequately established outcome for the task
Probability estimate	A numerical estimate of how likely a specified event is
Calibration	Agreement between probability estimates and observed frequencies across relevant groups of cases
Score meaning	The definition explaining what a reported number measures

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

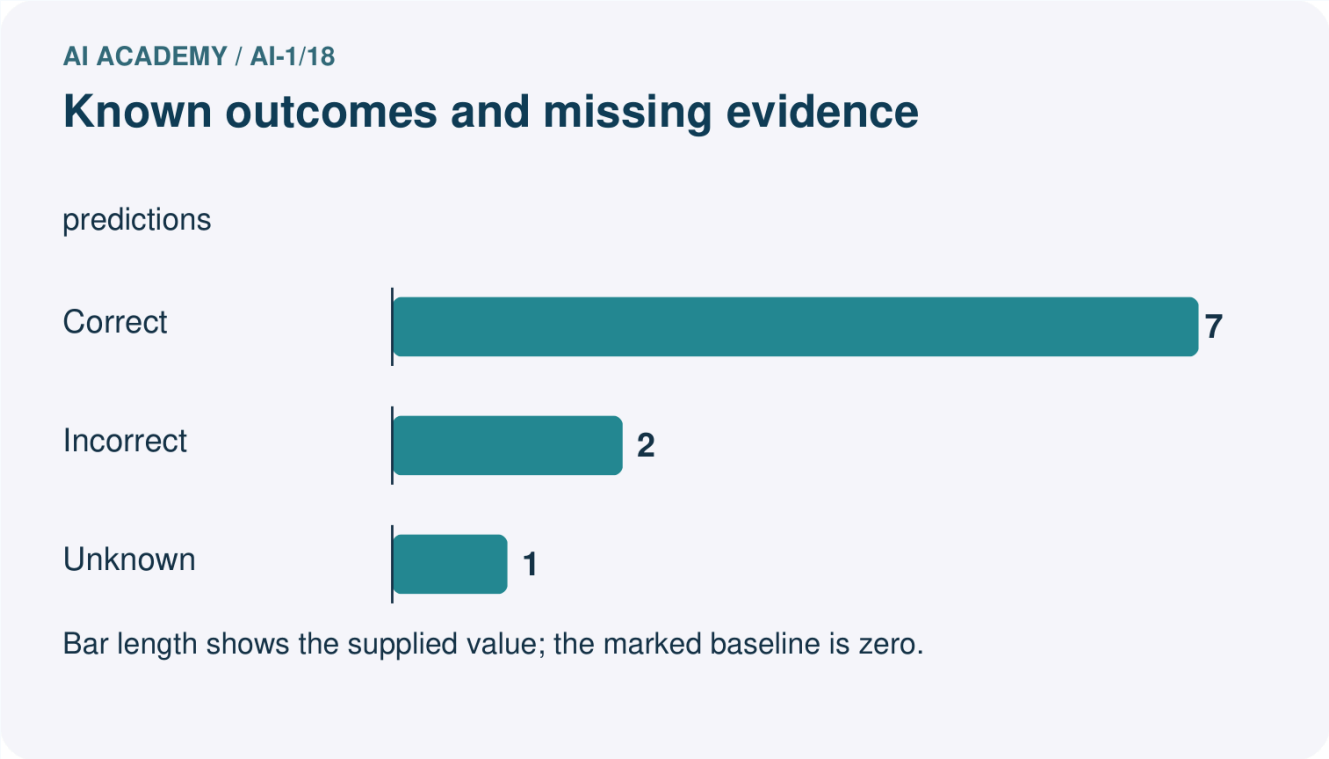
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Confidence Is Not Correctness

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/18

Confidence Is Not Correctness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep scores

Ten chosen labels each carry 80%; this is a claim about uncertainty.



2. Checked outcomes

Seven correct plus two wrong gives nine known outcomes.



3. Report coverage

Observed agreement is 7/9 among checked cards; outcome coverage is 9/10.



4. Retain unknown

The missing result could change the all-ten correct count to 7 or 8.

CHANGE ONE THING

The missing outcome is checked and found incorrect.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep scores	Ten chosen labels each carry 80%; this is a claim about uncertainty.
2. Checked outcomes	Seven correct plus two wrong gives nine known outcomes.
3. Report coverage	Observed agreement is 7/9 among checked cards; outcome coverage is 9/10.
4. Retain unknown	The missing result could change the all-ten correct count to 7 or 8.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Use cases A, B and C from the lesson table. State which is correct, incorrect and unassessed. Explain why ordering them by **confidence** cannot replace checking their outcomes.

Expected reasoning and fading help

Teacher check: A is incorrect, B correct and C unassessed. The score is a prior claim; the observed category determines agreement for an assessable case.

A larger claim does not supply a missing outcome or override a conflicting one.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Confidence** and **correctness**. Proposed explanation: “An 80 point score always means an 80 percent probability”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Read the documented quantity before interpreting it.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Was the unknown label guaranteed correct because its **confidence** was **80%**?

No. An **80%** score is not certainty. The new **evidence** records it as incorrect, and the complete agreement rate is **70%**.

Changed-case reasoning: All-ten agreement is 7/10=**70%**. There are now three wrong labels and no unknown outcomes.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A helper confidently labels a blurred dog photo as cat; the clear original shows a dog.

Write the question to explain: What does this show?

Reveal after **prediction**s: **Confidence** and **correctness** differ. A **confident error** remains wrong. Check **evidence**, particularly unfamiliar or low-quality **input**s.

Use the earlier changed question as optional later practice. Reserve Audit a different probability claim for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Each score claims an 80 percent chance that the chosen category is correct. Observed results are eight correct and two incorrect out of ten assessed.

This small group matches the claim numerically but cannot guarantee future performance or identify the next result.

Exit: confident language is not a defined probability, and repeated wording does not establish **correctness**; seek relevant outcome **evidence** and documented score evaluation.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve the need for an observed outcome before judging correctness.
7 to 20 minutes	Contrast a feature-match score, a defined probability estimate and confident wording. Explain each diagram row.
20 to 33 minutes	Complete W1 and W2. Require score meanings and outcome evidence in separate statements.
33 to 45 minutes	Use ten counters for W3. Compare the recorded claim with the observed fraction without turning the fraction into a guarantee.
Session 2 start 0 to 8 minutes	Retrieve why a high-confidence prediction can be wrong.
8 to 23 minutes	Complete the guided group report. Discuss the difference between numerical agreement in a small group and evidence about a wider method.

Stage	Teacher action
23 to 38 minutes	Collect W4 independently. Check that the missing outcome stays visible and the denominator is seven assessed cases.
38 to 45 minutes	Discuss W5 and the exit. End with the evidence needed for a meaningful confidence claim.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

In the supplied grid, A is incorrect despite strong wording; B is correct despite caution; C is correct on its single supplied check, which does not establish calibration; D is unassessed and its 80-point match score is not a probability. There are two correct predictions among three assessed records and one unassessed. Claim types differ, so avoid ranking them as though every display were the same probability measure. Reserve Audit a different probability claim for the individual first attempt.

Run and record

1. Complete the task: Compare **confident error**s with cautious correct answers.

2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

An 80 point score always means an 80 percent probability: Read the documented quantity before interpreting it.

An 80 percent method must succeed exactly eight times in every ten: A probability describes likelihood; small groups can vary.

A high-**confidence** error disproves all probabilities: Investigate many relevant cases while preserving individual errors.

Choose support from the response

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Audit a different probability claim for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

CLAIM TO REPAIR Very high **confidence** guarantees a correct label.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A missing outcome can be counted as a failure to finish the fraction: Report assessed results and missing counts separately.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: A fictional leaf sorter claims a 75 percent probability of **correctness** for each of eight chosen labels. Later **evidence** establishes six correct labels, one incorrect label and one missing outcome.

A report says six out of eight proves perfect **calibration** and guarantees the next label. Repair the report, state the assessed and unassessed counts, and describe a better follow-up check.

Seven cases are assessed: six correct and one incorrect; one is unassessed. Report six out of seven assessed alongside the missing count, and preserve the original 75 percent claims.

Six out of eight treats the missing outcome as known. The small incomplete group proves neither **calibration** nor the next label. Seek the missing outcome and evaluate the unchanged method on more relevant reserved cases.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Read the score definition

A card sorter gives a match score of 80 out of 100 points. Its documentation says the score counts weighted feature matches and is not a probability. A learner calls this an 80 percent chance of **correctness**.

Explain the error and write an accurate description.

The learner replaced the documented **score meaning** with a probability interpretation. Report 80 of 100 feature-match points.

To claim a chance of **correctness**, a defined probability method and relevant outcome evaluation would be needed.

W2 Judge cases using outcomes

Use cases A, B and C from the lesson table. State which is correct, incorrect and unassessed. Explain why ordering them by **confidence** cannot replace checking their outcomes.

A is incorrect, B correct and C unassessed. The score is a prior claim; the observed category determines agreement for an assessable case. A larger claim does not supply a missing outcome or override a conflicting one.

W3 Interpret a small group carefully

Ten **prediction**s each claim an 80 percent probability that their chosen category is correct. Five later prove correct and five incorrect. State the claimed probability and observed fraction separately.

Does this prove every future **prediction** will be wrong, or that each next **prediction** has exactly a 50 percent chance? Explain.

Claim 80 percent; observed five correct out of ten, or 50 percent in this group. This group falls below the claim but is small. It proves neither all future failures nor a precisely known 50 percent chance for each next case.

More relevant independent **evidence** is needed.

W5 Keep repetition separate from evidence

A learner copies the same 90 percent **prediction** into ten rows and calls this ten independent tests. No new outcomes were obtained. Explain why the **evidence** has not grown and propose a useful check.

The rows duplicate one **prediction**, not ten independently assessed cases. No new outcome **evidence** was obtained.

Preserve the original record, establish its outcome where possible, and gather distinct relevant reserved cases with their own **prediction**s and verified outcomes.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Confidence and **correctness** differ. A **confident error** remains wrong. Check **evidence**, particularly unfamiliar or low-quality **input**s.

Additional transfer practice question

A sorter is very confident that a leaf is a coin, but a person checks and sees a leaf. Which determines **correctness**: **confidence** or the checked object? Why?

Reasoned practice answer

The checked object determines **correctness**. **Confidence** describes the model's score or certainty, which can be misplaced. Keep the incorrect case as **evidence** for testing.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Very high **confidence** guarantees a correct label.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Both are 2/4 accurate because their answers are identical. Certainty language changes no outcome.

To compare **confidence** meaningfully, collect many **prediction**s with defined **confidence** categories or probabilities and compare their observed success rates; these four cases alone are insufficient.

Explanations for the additional workbook examples

W1 Read the score definition

W2 Judge cases using outcomes

W3 Interpret a small group carefully

W4 Audit a different probability claim

A fictional leaf sorter claims a 75 percent probability of **correctness** for each of eight chosen labels. Later **evidence** establishes six correct labels, one incorrect label and one missing outcome.

A report says six out of eight proves perfect **calibration** and guarantees the next label. Repair the report, state the assessed and unassessed counts, and describe a better follow-up check.

Seven cases are assessed: six correct and one incorrect; one is unassessed. Report six out of seven assessed alongside the missing count, and preserve the original 75 percent claims.

Six out of eight treats the missing outcome as known. The small incomplete group proves neither **calibration** nor the next label. Seek the missing outcome and evaluate the unchanged method on more relevant reserved cases.

W5 Keep repetition separate from evidence

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Record both the sorter's **confidence** wording and the human-confirmed outcome; keep them in separate columns.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Keep sorter scores, reference outcomes and missingness separately; use review policies instead of treating high scores as facts.

Support while preserving the reasoning goal

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Use counters and separate headings for claim and outcome. Read 80 percent as 80 out of 100 before comparing fractions.

Extension: explain why checking only easy cases might make the **confidence** method appear more dependable than it is.

W4 requires six of seven assessed, one unassessed, the original 75 percent claim, no **calibration** guarantee and a relevant follow-up.

After support retry ten 80 percent claims with seven correct, two incorrect and one missing: seven of nine assessed and one unassessed, with the same limits.

Plan the delayed check

Suggested return points in this level: ai-1/19, ai-1/21, ai-1/24. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, **confidence** rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. **Confidence** ratings are separate from correct independent performance.

Next: Week 19: Similarity. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 19 investigates how selected features change similarity and the examples a system considers close.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Helpful Classroom Sorter

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Distinguish a **confidence claim** from **evidence** that a **prediction** was correct

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Use cases A, B and C from the lesson table. State which is correct, incorrect and unassessed. Explain why ordering them by **confidence** cannot replace checking their outcomes.

Reasoned practice answer: A is incorrect, B correct and C unassessed. The score is a prior claim; the observed category determines agreement for an assessable case.

A larger claim does not supply a missing outcome or override a conflicting one.

Check every learner before discussion: Use the worked case for **Confidence** and **correctness**. Proposed explanation: “An 80 point score always means an 80 percent probability”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Read the documented quantity before interpreting it.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The missing outcome is checked and found incorrect. Require a **prediction** and a reason before comparing. Expected result: All-ten agreement is 7/10=**70%**.

Reason: There are now three wrong labels and no unknown outcomes.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from **correctness**.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from **correctness**, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

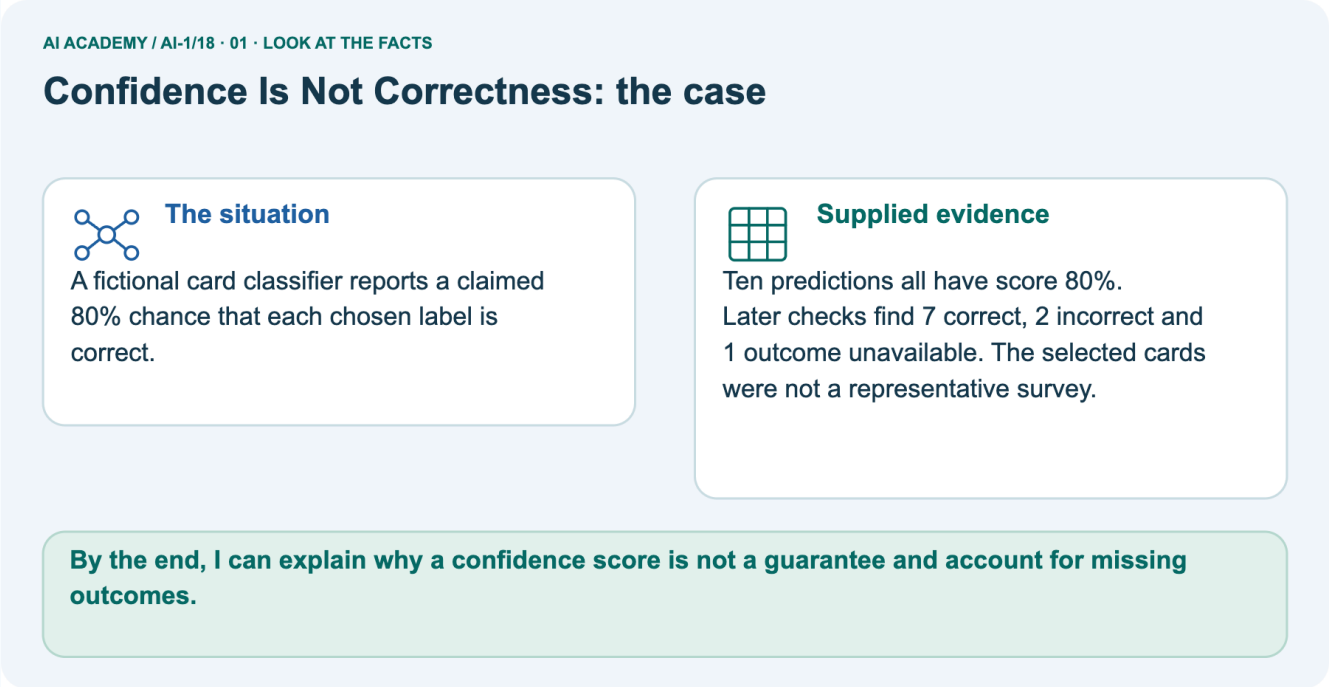
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Confidence Is Not Correctness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

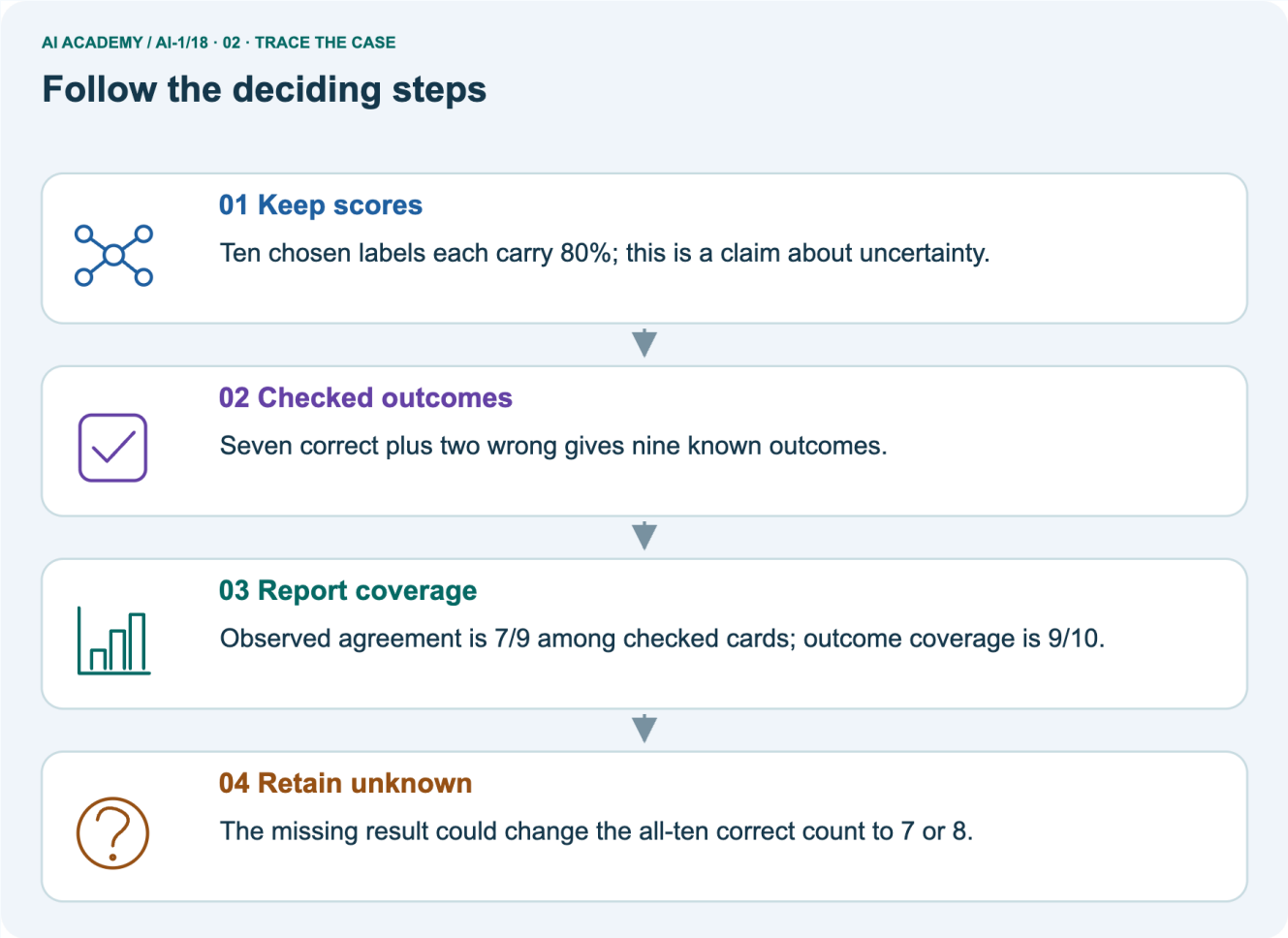
A fictional card classifier reports a claimed **80%** chance that each chosen label is correct. Supplied facts: Ten **prediction**s all have score **80%**. Later checks find 7 correct, 2 incorrect and 1 outcome unavailable.

The selected cards were not a representative survey.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep scores: Ten chosen labels each carry **80%**; this is a claim about uncertainty. → Checked outcomes: Seven correct plus two wrong gives nine known outcomes. → Report coverage: Observed agreement is 7/9 among checked cards; outcome coverage is 9/10. → Retain unknown: The missing result could change the all-ten correct count to 7 or 8.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Known outcomes and missing evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

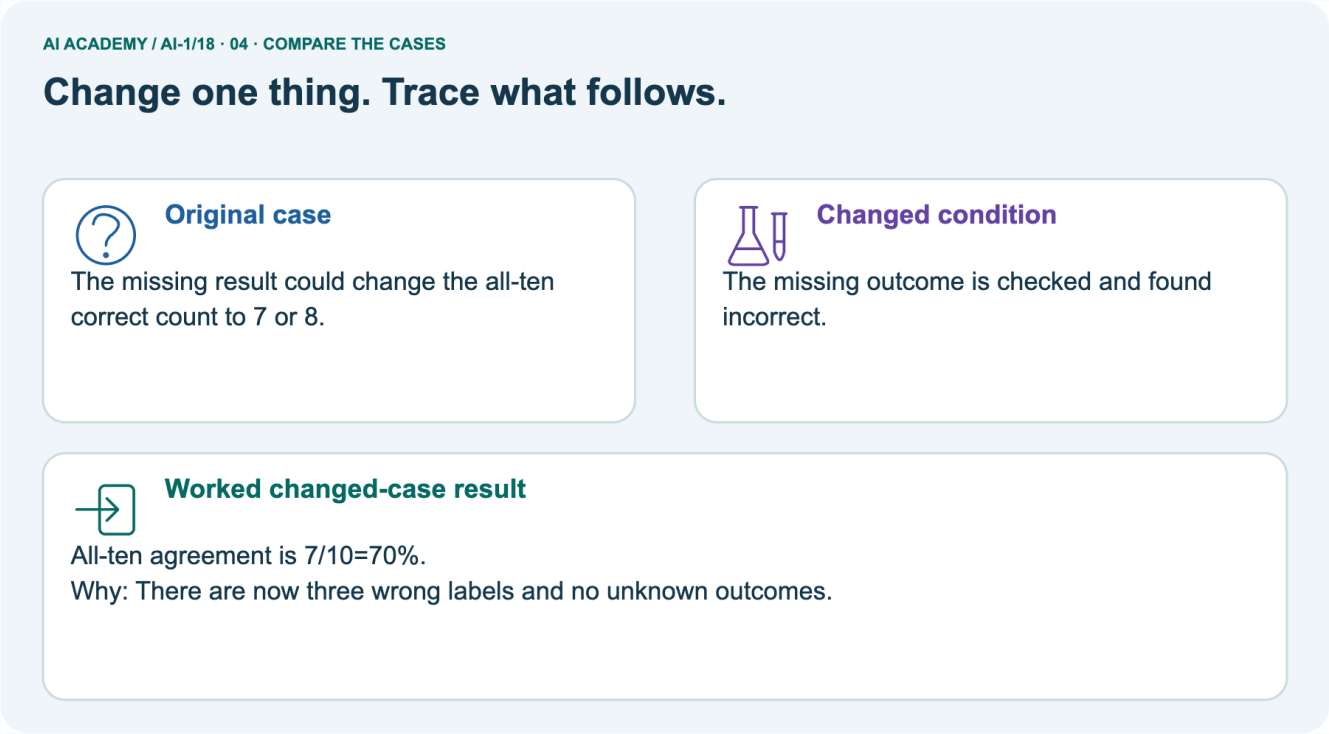
Known outcomes and missing **evidence**. Units: **prediction**s. Correct: 7. Incorrect: 2. Unknown: 1.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The missing result could change the all-ten correct count to 7 or 8. Changed condition: The missing outcome is checked and found incorrect. Expected result: All-ten agreement is 7/10=**70%**.

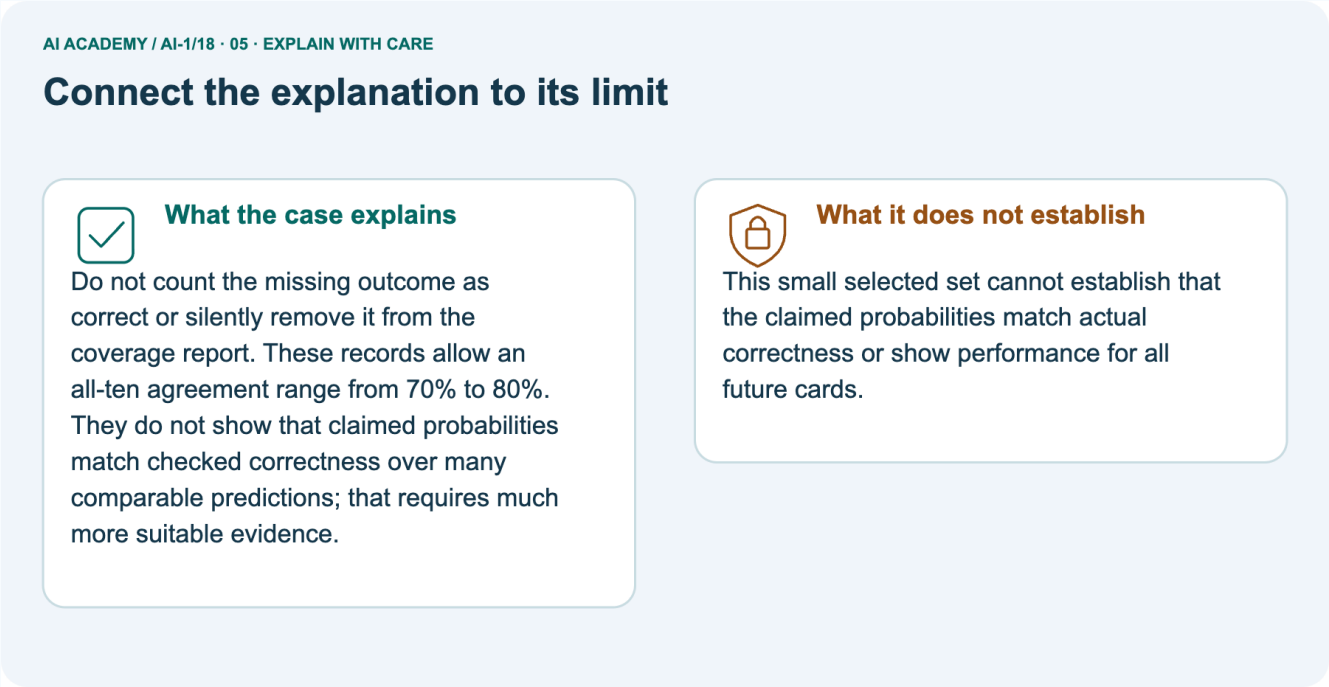
Reason: There are now three wrong labels and no unknown outcomes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: There are now three wrong labels and no unknown outcomes.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Do not count the missing outcome as correct or silently remove it from the coverage report. These records allow an all-ten agreement range from **70%** to **80%**.

They do not show that claimed probabilities match checked **correctness** over many comparable **prediction**s; that requires much more suitable **evidence**.

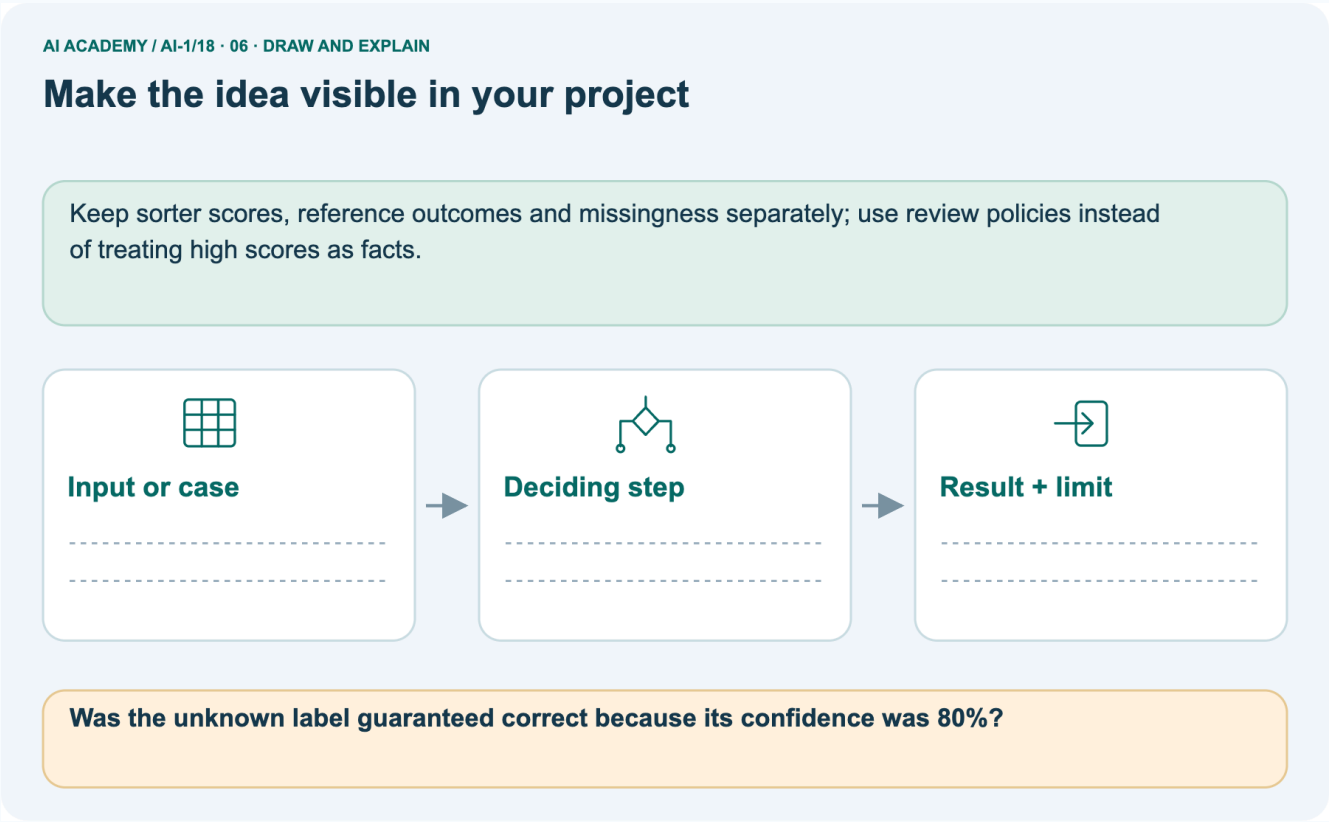
Limit: This small selected set cannot establish that the claimed probabilities match actual **correctness** or show performance for all future cards.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This small selected set cannot establish that the claimed probabilities match actual **correctness** or show performance for all future cards.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep sorter scores, reference outcomes and missingness separately; use review policies instead of treating high scores as facts. Draw your **input**, deciding step, result and limit.

Supported question: Was the unknown label guaranteed correct because its **confidence** was **80%**?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. An **80%** score is not certainty. The new **evidence** records it as incorrect, and the complete agreement rate is **70%**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Confidence Is Not Correctness: the case

02. Follow the deciding steps

03. Known outcomes and missing evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.